

TITLE PAGE

- **Problem Statement ID – SIH25032**
- **Problem Statement Title - Development of a smart digital platform to promote eco and cultural tourism in Jharkhand**
- **Theme - Travel & Tourism**
- **PS Category - Software**
- **Team ID – SRM-SIH-317**
- **Team Name (Registered on portal) – Ctrl Alt Hacks**



The project aims to build a smart, integrated web platform that showcases Jharkhand's eco-tourism attractions, cultural heritage, festivals, tribal traditions, and local businesses within a single digital ecosystem. The platform will act as a one-stop digital gateway for travelers while simultaneously empowering local communities and promoting sustainable tourism practices.

* Core Features



Trip Genie: Multilingual AI chatbot suggests personalized itineraries



Heritage Badge Quiz: Gamified learning + certificates



Elder-Friendly Mode: Big fonts + voice narration



Community Wall: Share experiences & photos (Users & locals)



Cultural Storytelling Audio: Interactive Audio narrations of heritage sites



Smart Weather & Time Suggestions: Visit the right place at the right time with real time notifications



Emergency Travel Assist: SOS button with helplines & maps



AR/VR Lens: 360° Immersive view of cultural sites & packages with interactive games



Innovation & Uniqueness

- AI-driven personalized cultural itineraries
- Gamified heritage learning with certificates & badges
- Community wall + local integration for sustainable tourism
- Inclusive design: regional languages + elder-friendly mode
- Immersive audio tales & games for sites & traditions



Govt Analytics – Data shared with the tourism board for insights & growth



Partnership & Scalability – Collaboration with local business & artisans



Sentiment Analysis – Real time feedback analysis & AI-driven dashboard with advice for the government

Our solution directly tackles this by creating a centralized digital platform that combines AI-powered personalized recommendations, real-time travel assistance, and cultural storytelling. It not only makes authentic information easily accessible to tourists but also integrates local artisans, communities, and agencies, fostering inclusive, sustainable, and culturally rich tourism in Jharkhand.

Tech Stack

 Frontend: Next.js, ShadCN UI, Tailwind CSS

 Backend: Node.js + Express/Spring Boot

 Database: Supabase (PostgreSQL)

 Authentication: Clerk

 AI: LLaMA 3 (Hugging Face)

 APIs: OpenWeather, Puppeteer

System Architecture Benefits

Scalability: Cloud-based architecture enables handling peak tourist seasons

Security: Clerk handles authentication, Supabase manages data protection

Performance: Next.js with server components optimizes loading speed

Integration: APIs enable weather forecasts, emergency services, and local business connections



System Flow

- 1 User logs in → Dashboard created
- 2 Trip Genie chatbot → asks preferences → suggests itineraries
- 3 Journey Hub → browse travel packages & guides
- 4 Heritage Badge → quizzes → auto-certificates
- 5 Gallery + Community Wall → cultural engagement & local experiences
- 6 Emergency Travel Assist → SOS + helplines
- 7 Agency Dashboard → insights for package optimization



SHADCN/UI



NEXTJS 14



SUPABASE

✔ Feasibility Assessment

</> Open-source & Cloud Tools
Built with cost-effective technologies that scale efficiently with user demand. Leverages modern web standards and cloud infrastructure.

🤖 LLaMA 3 Implementation
Fast, scalable AI inference for personalized cultural recommendations and trip planning assistance.

🛡️ Supabase + Clerk Integration
Secure user management and authentication system with robust database capabilities.

⚠️ Risks & Mitigation Strategies

⚡ AI & API Limits
High traffic could cause AI service disruptions and increase API costs.

🛡️ Mitigation
Implement intelligent caching system for frequent queries and response patterns.

🗉 Language Barrier
Limited accessibility for non-Hindi/English speaking users, especially tribal communities.

🛡️ Mitigation
Add bilingual support including Santali and other regional languages.

! Cultural Data Accuracy
Incorrect cultural information could lead to misrepresentation of heritage.

🛡️ Mitigation
Collaborate with Govt. of Jharkhand & NGOs to verify cultural information.

Viability 🌐

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Supports tourists, agencies, and local communities → ensures adoption.
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Aligns with Govt. of Jharkhand tourism goals → strong chance for partnerships.
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Revenue potential via agency analytics & local community onboarding.
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Scalable to other states → long-term sustainability.

Monitoring & Quality Assurance

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Real-time performance monitoring
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Community feedback loops
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Regular content audits
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Continuous integration

IMPACT AND BENEFITS



Impact on Stakeholders



Tourists

- Personalized cultural itineraries based on preferences
- Gamified learning about heritage and traditions
- Authentic cultural exploration experiences
- Enhanced safety with emergency travel assistance



Local Communities

- Increased visibility for artisans and performers
- Direct promotion of local guides and services
- Preservation of tribal traditions and knowledge
- Platform for sharing authentic cultural stories



Agencies & Government

- Data-driven insights for tourism planning
- Better promotion of heritage tourism sites
- Analytics for optimizing tourism infrastructure
- Digital platform for cultural preservation

Sustainable Benefits



Cultural Benefits

Gamified heritage education, digital preservation of traditions, and authentic representation of Jharkhand's diverse cultural landscape.



Social Benefits

Inclusive design with elder-friendly features, multilingual support including regional languages, and accessibility options for diverse users.



Economic Benefits

Boosts local tourism economy, creates digital marketplace for artisans, enhances visibility for small businesses, and facilitates tourism partnerships.



Environmental Benefits

Promotes eco-friendly travel options, encourages community participation in conservation, and highlights sustainable tourism practices.



ATTRactions AND
EXPERIENCES



RETAIL AND SMALL TO
MEDIUM ENTERPRISE (SME)



ACCOMMODATION AND
TOURISM TRANSPORTATION

Government & Tourism

 [Govt. of Jharkhand Tourism Portal](#)
Official tourism statistics and cultural heritage data

 [Ministry of Tourism, Govt. of India](#)
National tourism policies and cultural guidelines

 [Jharkhand Tribal Research Institute](#)
Cultural authenticity and tribal heritage verification

Technical Documentation

 [Hugging Face LLaMA 3](#)
AI integration for personalized tourism recommendations

 [Supabase & Clerk Official Documentation](#)
Database management and authentication implementation

 [OpenWeather API Documentation](#)
Weather data integration for tourism recommendations

Research & Case Studies

 [Case Studies on AI in Heritage Tourism](#)
Academic research on AI applications in cultural tourism

 [Digital Transformation in Tourism Industry](#)
Analysis of technology impact on tourism experiences

 [Sustainable Tourism Impact Assessment](#)
Measuring economic & social benefits of sustainable tourism